

Karan Shihire

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EDUCATION

Modern College of Arts, Science and Commerce

Bachelor of Computer Science (Minor: Embedded Systems & IoT) | CGPA: 8.0

Pune, Maharashtra

Jul 2023 – May 2026

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate

Cloud Architecture, OCI Core Services, Identity & Security

Oracle

PROJECTS

- **AI-Powered Real-Time Face Recognition Attendance & Surveillance System** | *Python, React, FastAPI, PostgreSQL, I*
 - Built a real-time face recognition system capable of detecting and identifying multiple faces from live webcam streams using InsightFace and OpenCV.
 - Implemented automated face enrollment pipeline with multi-angle image capture, face alignment, embedding extraction, and vector storage for reliable recognition.
 - Developed embedding similarity search workflow using FAISS to perform fast identity matching on high-dimensional facial embeddings.
 - Integrated ONNX Runtime inference pipeline to optimize face recognition performance and reduce detection latency during live video processing.
 - Engineered PostgreSQL database schema for storing user profiles, facial embeddings metadata, attendance logs, and surveillance event records.
 - Developed full-stack AI application integrating computer vision models, backend APIs, database systems, and responsive frontend interfaces.
- **AI-Powered Network Intrusion Detection System (IDS)** | *Python, Scapy, Isolation Forest, Flask* [GitHub](#)
 - Developed an AI-driven intrusion detection system using Isolation Forest for anomaly detection on live network traffic data.
 - Implemented packet capture and traffic analysis pipeline using Scapy for TCP/UDP/ICMP monitoring and suspicious activity detection.
 - Built a real-time Flask dashboard for visualizing packet statistics, anomaly alerts, and live monitoring data.
 - Designed concurrent detection workflow for feature extraction, ML inference, and structured alert generation on real-time traffic streams.
- **QuizNet — AI-Powered Quiz Generator** | *Python, Django, Google Gemini API, SQLite* [GitHub](#)
 - Integrated Google Gemini LLM API to dynamically generate MCQ, reasoning, and true/false questions using prompt engineering workflows.
 - Built Django backend with session management, scoring logic, and interactive UI for AI-generated quiz delivery.

TECHNICAL SKILLS

Programming: Python, SQL, Java

AI / Machine Learning: Computer Vision, Deep Learning, Facial Recognition, Embedding Generation, Real-Time AI Inference, Image Processing

Frameworks & Libraries: OpenCV, PyTorch (Basics), InsightFace, ONNX Runtime, FAISS, NumPy, Pandas, Scikit-learn, FastAPI, Django, Flask

Web & Backend: React.js, REST APIs, WebSocket, PostgreSQL

Tools & Platforms: Git, GitHub, VS Code, Cursor, Oracle OCI, Render, Netlify